

姓名:

学院:

学号:

上海科技大学

2023-2024 学年第 2 学期本科生期末考试卷

开课单位：信息学院

授课教师：邵子瑜

考试科目：《面向信息科学的概率论与数理统计》

课程代码：SI 140A

考试时间：2024 年 6 月 25 日 9 点 00 分 -12 点 00 分。

考试成绩录入表：

题目	1	2	3	4	5	6	7	8	9	总分
计分										
复核										

评卷人签名：复核人签名：

日期：日期：

编写说明：

1. 要求评卷人和复核人不能是同一人。
2. 试卷内页和答题纸编排格式由各学院和出题教师根据实际需要自定，每页须按顺序标注页码(除封面外)，要求排版清晰、美观，便于在页面左侧装订。为方便印刷归档，建议使用 A4 双面印刷（学校有印刷一体机提供）。
3. 主考教师编写试卷时尽可能保证试题科学、准确、合理，如考试过程中发现试题有误，主考教师需负责现场解释，此类情况学校将作为教学评估记录的一部分。

Probability & Statistics for EECS
Spring 2024

Final
2024/06/25

Time Limit: 180 Minutes

Name (Print): _____

Advisor Name _____

This exam contains 11 pages (including this cover page) and 9 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

Try to answer as many problems as you can. The following rules apply:

- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit.

Do not write in the table to the right.

Problem	Points	Score
1	20	
2	20	
3	20	
4	5	
5	5	
6	5	
7	5	
8	10	
9	10	
Total:	100	

1. (20 points) Given two i.i.d. random variables X and Y satisfying distribution $\text{Unif}(0, 5)$.
 - (a) (10 points) Find $P(\max(X, Y) \leq 1)$.
 - (b) (10 points) Find $P(\min(X, Y) \leq 1)$.

2. (20 points) Let X and Y be two continuous random variables with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} c & \text{if } 0 < x < 1, 0 < y < 2x, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) (10 points) Find the value of constant c .
- (b) (5 points) Find the marginal distribution of X .
- (c) (5 points) Find the marginal distribution of Y .

3. (20 points) Given a random variable $X \sim \text{Expo}(\lambda)$, where $\lambda > 0$.
- (a) (10 points) Find $E(X|X > 2030)$.
 - (b) (10 points) Find $E(X|X < 2030)$.

4. (5 points) Let $X \sim \text{Pois}(\lambda)$, where $\lambda > 0$. Find $E\left[\binom{X}{k}\right]$, where k is a nonnegative integer.

5. (5 points) U, V, W are three independent random variables satisfying the uniform distribution $\text{Unif}(0, 1)$. Let $S = U + V + W$. Find the value of $E(\text{Var}(U|S))$.

6. (5 points) Let $X \sim \text{Unif}(0, 1)$, and express X as a continued fraction in the following format:

$$X = \frac{1}{Y_1 + \frac{1}{Y_2 + \frac{1}{Y_3 + \dots}}}$$

Find the joint distribution of Y_1 and Y_2 .

7. (5 points) Given a random variable X satisfying exponential distribution, i.e., $X \sim \text{Expo}(\lambda)$, where $\lambda > 0$. Let $Y = \lfloor X \rfloor$ be the integer part of X , and $Z = X - Y$ be the fractional part of X . Find the joint distribution of Y and Z .

8. (10 points) Let $X_1, \dots, X_n$ be i.i.d. r.v.s with mean μ and variance σ^2 , and $n \geq 2$. A bootstrap sample of $X_1, \dots, X_n$ is a sample of n r.v.s $X_1^*, \dots, X_n^*$ formed from the $X_j, \forall j \in \{1, \dots, n\}$ by sampling with replacement with equal probabilities. Let $\bar{X}^*$ denote the sample mean of the bootstrap sample:

$$\bar{X}^* = \frac{1}{n}(X_1^* + \dots + X_n^*).$$

- (a) (5 points) Calculate $E(X_j^*)$ and $\text{Var}(X_j^*)$ for each $j \in \{1, \dots, n\}$.
- (b) (5 points) Calculate $E(\bar{X}^*)$ and $\text{Var}(\bar{X}^*)$. Then explain intuitively why $\text{Var}(\bar{X}) < \text{Var}(\bar{X}^*)$.

9. (10 points) There are two urns with a total of $2N$ distinguishable balls. Initially, the first urn has N white balls and the second urn has N black balls. At each stage, we pick a ball at random from each urn and interchange them. Let X_n be the number of black balls in the first urn at time n . This is a Markov chain on the state space $\{0, 1, \dots, N\}$.
- (a) (5 points) Find the transition probabilities of the chain.
 - (b) (5 points) Find the stationary distribution of the chain.

	Y discrete	Y continuous
X discrete	$P(Y = y X = x) = \frac{P(X=x Y=y)P(Y=y)}{P(X=x)}$	$f_Y(y X = x) = \frac{P(X=x Y=y)f_Y(y)}{P(X=x)}$
X continuous	$P(Y = y X = x) = \frac{f_X(x Y=y)P(Y=y)}{f_X(x)}$	$f_{Y X}(y x) = \frac{f_{X Y}(x y)f_Y(y)}{f_X(x)}$

	Y discrete	Y continuous
X discrete	$P(X = x) = \sum_y P(X = x Y = y)P(Y = y)$	$P(X = x) = \int_{-\infty}^{\infty} P(X = x Y = y)f_Y(y)dy$
X continuous	$f_X(x) = \sum_y f_X(x Y = y)P(Y = y)$	$f_X(x) = \int_{-\infty}^{\infty} f_{X Y}(x y)f_Y(y)dy$

Figure 1: Bayes' Rule & LOTP.

Table of distributions

Name	Param.	PMF or PDF	Mean	Variance
Bernoulli	p	$P(X = 1) = p, P(X = 0) = q$	p	pq
Binomial	n, p	$\binom{n}{k} p^k q^{n-k}$, for $k \in \{0, 1, \dots, n\}$	np	npq
FS	p	pq^{k-1} , for $k \in \{1, 2, \dots\}$	$1/p$	q/p^2
Geom	p	pq^k , for $k \in \{0, 1, 2, \dots\}$	q/p	q/p^2
NBinom	r, p	$\binom{r+n-1}{r-1} p^r q^n$, $n \in \{0, 1, 2, \dots\}$	rq/p	rq/p^2
HGeom	w, b, n	$\frac{\binom{n}{k} \binom{w+b-k}{n-k}}{\binom{w+b}{n}}$, for $k \in \{0, 1, \dots, n\}$	$\mu = \frac{nw}{w+b}$	$(\frac{w+b-n}{w+b-1})n\frac{w}{w+b}(1 - \frac{w}{w+b})$
Poisson	λ	$\frac{e^{-\lambda} \lambda^k}{k!}$, for $k \in \{0, 1, 2, \dots\}$	λ	λ
Uniform	$a < b$	$\frac{1}{b-a}$, for $x \in (a, b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	μ, σ^2	$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2
Log-Normal	μ, σ^2	$\frac{1}{x\sigma\sqrt{2\pi}} e^{-(\ln x - \mu)^2/(2\sigma^2)}$, $x > 0$	$\theta = e^{\mu + \sigma^2/2}$	$\theta^2(e^{\sigma^2} - 1)$
Expo	λ	$\lambda e^{-\lambda x}$, for $x > 0$	$1/\lambda$	$1/\lambda^2$
Gamma	a, λ	$\Gamma(a)^{-1} (\lambda x)^a e^{-\lambda x} x^{-1}$, for $x > 0$	a/λ	a/λ^2
Beta	a, b	$\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}$, for $0 < x < 1$	$\mu = \frac{a}{a+b}$	$\frac{a(1-\mu)}{a+b+1}$
Chi-Square	n	$\frac{1}{2^{n/2} \Gamma(n/2)} x^{n/2-1} e^{-x/2}$, for $x > 0$	n	$2n$
Student-t	n	$\frac{\Gamma((n+1)/2)}{\sqrt{n\pi} \Gamma(n/2)} (1 + x^2/n)^{-(n+1)/2}$	0 if $n > 1$	$\frac{n}{n-2}$ if $n > 2$